2026 Wrestling Championships

Predicted Champions Summary

Machine Learning Model Predictions · 4 Models · 3 Conferences · 10 Weight Classes Each

4	3	10	248+
ML Models	Conferences	Weight Classes Each	Total Matchups

★ **XGBoost 2 is the BEST MODEL** — it adds **team_rank_diff** to capture program quality, recruitment depth, and coaching strength that individual stats alone cannot reveal.

THE PREDICTION MODELS

Four machine learning algorithms — each with different logic, features, and depth

LR

MODEL 1

BASELINE MODEL · 1 FEATURE

Logistic Regression

What it is: Logistic Regression is a classical statistical model that predicts the probability of a binary outcome — in this case, Win or Loss. Despite the name, it is a *classification* algorithm, not a regression one. It works by fitting a weighted linear combination of input features, then passing the result through a **sigmoid function** (S-shaped curve) that maps any real number to a value between 0 and 1. That output is interpreted directly as a win probability.

How it works here: The model learns a single weight for **win_rate_diff** (home wrestler's win rate minus away wrestler's win rate). During training, it adjusts this weight to minimize prediction error across historical matchups using **maximum likelihood estimation** — finding the parameter that makes the observed outcomes most probable.

Strengths: Extremely interpretable — you can directly read the learned coefficient to understand how much win rate difference affects win probability. Fast to train, resistant to overfitting on small datasets, and provides well-calibrated probabilities.

Limitation: Only uses one feature. Cannot capture non-linear patterns, interaction effects, or contextual factors like program quality or rematch history.

Features:

```
1 feature: win_rate_diff (home win rate minus away win rate)
```

DT

MODEL 2

33 FEATURES · INTERPRETABLE

Decision Tree

What it is: A Decision Tree is a non-parametric supervised learning algorithm that partitions the feature space into rectangular regions through a series of hierarchical if/else splits. Each internal node tests one feature against a threshold; each leaf node assigns a prediction. The result is a human-readable flowchart of decision logic.

How it works here: At each node, the tree selects the feature and split point that maximizes **information gain** (or minimizes Gini impurity) — i.e., the split that best separates wins from losses in the training data. It recurses on each sub-partition until a stopping criterion is met (max depth, min samples per leaf, etc.). To predict a matchup, the model routes the feature vector down the tree from root to leaf and returns the majority class of training samples that landed in that leaf.

Strengths: Naturally handles non-linear relationships and feature interactions without any transformations. You can inspect the tree to see exactly why a prediction was made — e.g., 'if win_rate_diff > 0.15 AND pin_count_diff > 2, predict home win.' Handles mixed data types natively.

Limitation: Prone to overfitting on training data without careful pruning. A single tree can be unstable — small changes in data can produce very different trees. This is why ensemble methods like XGBoost were developed.

Features:

```
33 features: wrestled_before · home_point_diff_rematches · home_pinned_away ·
matches_wrestled_diff · win_rate_diff · bonus_rate_diff · pin_count_diff ·
avg_point_diff_diff · avg_points_scored_diff · std_point_diff_diff · weight_class
encodings (×10) · home academic class (×7) · away academic class (×7)
```

XGB1

MODEL 3

33 FEATURES · GRADIENT BOOSTING

XGBoost v1

What it is: XGBoost (Extreme Gradient Boosting) is an ensemble learning method that builds hundreds of decision trees *sequentially*, where each new tree is trained to correct the residual errors of the combined ensemble so far. This process — called **gradient boosting** — optimizes a differentiable loss function (log-loss for classification) using gradient descent in function space.

How it works here: Starting from an initial prediction (base score), each successive tree fits the **negative gradient** of the loss — essentially the direction of steepest improvement. A **learning rate** (shrinkage) controls how much each tree contributes, preventing any single tree from dominating. XGBoost also applies **L1 and L2 regularization** on tree weights and leaf scores, and uses **second-order Taylor expansion** of the loss function (using both gradient and Hessian) to make more precise updates than standard gradient boosting.

Strengths: Extremely high predictive accuracy. Handles missing data natively via learned default directions at each split. Built-in regularization prevents overfitting. Captures complex non-linear interactions across all 33 features simultaneously — interactions that Logistic Regression and a single Decision Tree cannot model.

Same features as Decision Tree — the key advantage is the ensemble of hundreds of trees, not additional inputs.

Features:

```
33 features (identical to Decision Tree above)
```

XGB2

MODEL 4

34 FEATURES · BEST MODEL ★

XGBoost v2

What it is: Same XGBoost algorithm as Model 3 — gradient-boosted decision trees with L1/L2 regularization and second-order gradient updates — but with one critical additional feature that qualitatively changes what the model can see: **team_rank_diff**.

What team_rank_diff captures: The difference in national program rankings between the two wrestlers' teams. This single number encodes a wealth of contextual signal invisible to individual statistics: the quality of daily training partners, coaching staff depth, recruiting resources, weight room facilities, nutrition support, travel budget, and the culture of winning in a program. A wrestler from the #1-ranked program who has a 75% win rate has almost certainly been tested by elite competition every day in practice — while a wrestler with the same win rate from a #50 program may have compiled that record against weaker opponents.

Why this makes it the best model: Wrestling outcomes are shaped not just by individual statistics but by the *environment* in which a wrestler was developed. team_rank_diff is the only feature in any of the four models that captures this program-level quality signal. In testing, adding this feature consistently improved prediction accuracy — which is why XGBoost v2 is designated the best model and its champion predictions are highlighted throughout this report.

Features:

```
All 33 DT/XGB1 features PLUS: team_rank_diff ← UNIQUE TO THIS MODEL
```

HOW TO INTERPRET THE AGREEMENT %

Understanding model consensus across all four algorithms

Agreement % Formula: (Matchups where ALL 4 models predict the same winner) ÷ (Total matchups) × 100

Example: In the Big Ten 174 lbs weight class, there are 91 matchups. All 4 models agreed on 79 of them → 79/91 × 100 = **86.8% agreement**.

Agreement %	Confidence Level	What It Means
90 – 100%	■ Very High	Models strongly agree — predictions most reliable
75 – 89%	■ High	A few close matchups, but generally strong agreement
60 – 74%	■ Moderate	Several contested matchups — upsets are plausible
Below 60%	■ Low	Many unpredictable matchups — highly competitive weight class

Symbol	Meaning
■ Unanimous	All 4 models predict the same champion for that weight class
■■ Split	At least one model disagrees on the champion prediction

★ XGB v2	XGBoost v2 column is highlighted — this is the BEST MODEL (includes team_rank_diff)

Big Ten Wrestling Championship 2026

10 Weight Classes · 91 Matchups per Class · 910 Total Matchups

Weight	LogReg	Dec. Tree	XGBoost 1	XGBoost 2 ★	Consensus	Agree %
125 lbs	Luke Lilledahl	Luke Lilledahl	Luke Lilledahl	**Luke Lilledahl**	■ Unanimous	84.6%
133 lbs	Marcus Blaze	Marcus Blaze	Marcus Blaze	**Marcus Blaze**	■ Unanimous	76.9%
141 lbs	Jesse Mendez	Jesse Mendez	Jesse Mendez	**Jesse Mendez**	■ Unanimous	81.3%
149 lbs	Shayne Van Ness	Shayne Van Ness	Shayne Van Ness	**Shayne Van Ness**	■ Unanimous	79.1%
157 lbs	Brandon Cannon	Brandon Cannon	Brandon Cannon	**Brandon Cannon**	■ Unanimous	85.7%
165 lbs	M. Mesenbrink	M. Mesenbrink	M. Mesenbrink	**M. Mesenbrink**	■ Unanimous	85.7%
174 lbs	Levi Haines	Levi Haines	Levi Haines	**Levi Haines**	■ Unanimous	86.8%
184 lbs	Rocco Welsh	Max McEnelly	Rocco Welsh	**Rocco Welsh**	■■ Split	84.6%
197 lbs	Josh Barr	Josh Barr	Josh Barr	**Josh Barr**	■ Unanimous	70.3%
285 lbs	Taye Ghadiali	Taye Ghadiali	Taye Ghadiali	**AJ Ferrari**	■■ Split	85.7%

★ XGBoost 2 = best model. **Green % ≥80% Amber % 65–79% Red % <65%**

Ivy League Wrestling Championship 2026

10 Weight Classes · 15 Matchups per Class · 150 Total Matchups

Weight	LogReg	Dec. Tree	XGBoost 1	XGBoost 2 ★	Consensus	Agree %
125 lbs	M-A McGowan	M-A McGowan	M-A McGowan	**M-A McGowan**	■ Unanimous	80.0%
133 lbs	Evan Mougalian	Evan Mougalian	Evin Gursoy	**Evin Gursoy**	■■ Split	86.7%
141 lbs	Vince Cornella	CJ Composto	Vince Cornella	**Vince Cornella**	■■ Split	86.7%
149 lbs	Jaxon Joy	Jaxon Joy	Jaxon Joy	**Jaxon Joy**	■ Unanimous	66.7%

Weight	LogReg	Dec. Tree	XGBoost 1	XGBoost 2 ★	Consensus	Agree %
157 lbs	Meyer Shapiro	Meyer Shapiro	Meyer Shapiro	**Meyer Shapiro**	■ Unanimous	80.0%
165 lbs	Cesar Alvan	Ty Whalen	Cesar Alvan	**Cesar Alvan**	■■ Split	66.7%
174 lbs	Simon Ruiz	Simon Ruiz	Nick Fine	**Nick Fine**	■■ Split	73.3%
184 lbs	Joe Curtis	Joe Curtis	Joe Curtis	**Joe Curtis**	■ Unanimous	73.3%
197 lbs	Andrew Reall	Andrew Reall	Andrew Reall	**Jack Wehmeyer**	■■ Split	73.3%
285 lbs	Alex Semenenko	Vincent Mueller	Vincent Mueller	**Vincent Mueller**	■■ Split	66.7%

★ XGBoost 2 = best model. **Green % ≥80% Amber % 65–79% Red % <65%**

ACC Wrestling Championship 2026

10 Weight Classes · 21 Matchups per Class · 210 Total Matchups

Weight	LogReg	Dec. Tree	XGBoost 1	XGBoost 2 ★	Consensus	Agree %
125 lbs	Eddie Ventresca	Eddie Ventresca	Eddie Ventresca	**Eddie Ventresca**	■ Unanimous	85.7%
133 lbs	Tyler Knox	Aaron Seidel	Aaron Seidel	**Aaron Seidel**	■■ Split	85.7%
141 lbs	Gable Porter	Gable Porter	Ryan Jack	**Ryan Jack**	■■ Split	57.1%
149 lbs	Koy Buesgens	Koy Buesgens	Koy Buesgens	**Koy Buesgens**	■ Unanimous	85.7%
157 lbs	Daniel Cardenas	Daniel Cardenas	Daniel Cardenas	**Daniel Cardenas**	■ Unanimous	90.5%
165 lbs	Will Denny	Will Denny	Will Denny	**Will Denny**	■ Unanimous	81.0%
174 lbs	Aiden Wallace	Matthew Singleton	Matthew Singleton	**Matthew Singleton**	■■ Split	71.4%
184 lbs	Jake Dailey	Jake Dailey	Jake Dailey	**Jaden Bullock**	■■ Split	66.7%
197 lbs	Sonny Sasso	Mac Stout	Sonny Sasso	**Angelo Posada**	■■ Split	76.2%
285 lbs	Isaac Trumble	Isaac Trumble	Isaac Trumble	**Isaac Trumble**	■ Unanimous	76.2%

★ XGBoost 2 = best model. **Green % ≥80% Amber % 65–79% Red % <65%**